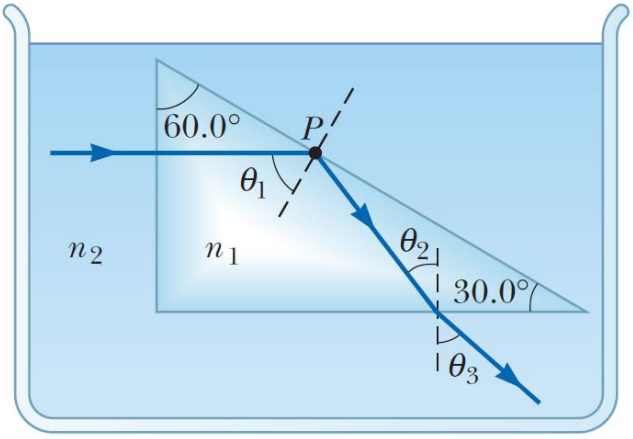
Assignment 1

1) As shown in figure, a light ray is incident normal to one face of a $30°-60°-90°$ prism of flint glass $(n = 1.66)$ that is immersed in water.

   a) Determine the exit angle θ_3 of the ray.

   b) A substance is dissolved in the water to increase the index of refraction n_2. At what value of n_2 does total internal reflection cease at point P ?



2) Three polarizing filters are stacked, with the polarizing axis of the second and third filters at $23.0°$ and $62.0°$, respectively, to that of the first. If unpolarized light is incident on the stack, the light has intensity $55.0\,\frac{W}{cm^2}$ after it passes through the stack. If the incident intensity is kept constant but the second polarizer is removed, what is the intensity of the light after it has passed

   through the stack?

3) The refractive index of a certain glass is 1.66. For what incident angle is light reflected from the surface of this glass completely polarized if the glass is immersed in (a) air and (b) water $(n = 1.33)$?

4) The index of refraction for violet light in silica flint glass is 1.66 and that for red light is 1.62. What is the angular separation of visible light passing through an equilateral prism of apex angle 60.0° if the angle of incidence is 50.0°?